FCT SERVICE NOTE
SN06082812

Class A Service information enclosed.

Date: August 28th, 2006

Subject: FRU Substitution for A2K1Contactor Used in the Compact Power Distribution Unit (CPDU)

Effectivity: All systems utilizing the CPDUs 2335370, 2269902 & 2133533
PET - Discovery ST (non-Hpower) & Discovery LS
CT - CT/i & LightSpeed (1x-4x) systems

Details:

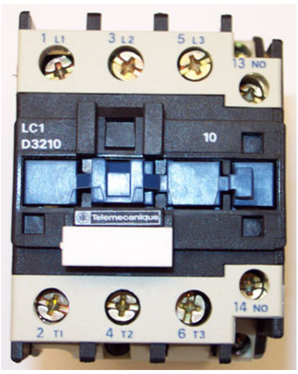
The OEM of the Telemecanique Contactor LC1D3210G6 (GEHC # 2133533-4) is no longer supplying this part. A substitute part, Telemecanique/ Square D Contactor LC1D32G7 (GEHC # 2269902-11), has been selected, which has similar specifications but minor Form Fit and Function differences. This new contactor has been purchased and stocked by Global Parts and Repair Services (GP&RS). Confusion has occurred on the new part's compatibility.

Resolution:

This Service Note will explain the differences and give guidance on the new contactor in the CPDU application.

Original Contactor 2133533-4

FEMC Description: Contactor, Telemecanique 110/50-60HZ



Model # LC1D3210G6

Specifications:
- 3 Pole, 600V
- I_{FLA} 32A Inductive / 50A Resistive
- 110V, 50/60 Hz Coil Voltage
- 1 Aux. Contacts (1 NO)

New Contactor 2269902-11

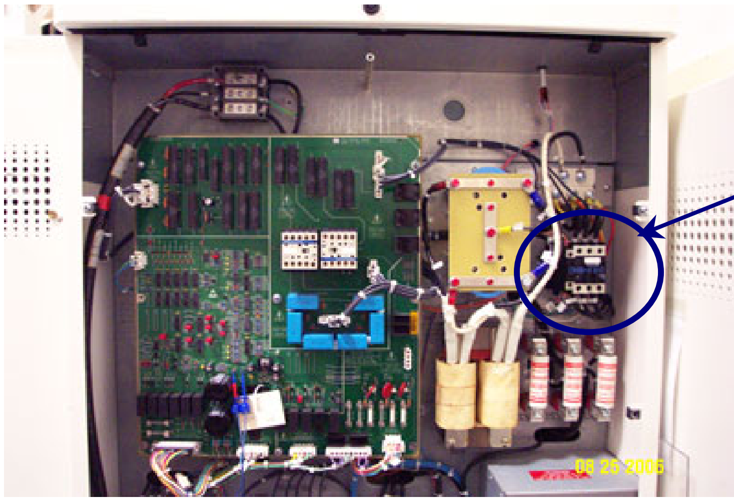
FEMC Description: Contactor, 32/50A, 3P, 120Vac Coil

Model # LC1D32G7

Specifications:
- 3 Pole, 600V
- I_{FLA} 32A Inductive / 50A Resistive
- 120V, 50/60 Hz Coil Voltage
- 2 Aux. Contacts (1 NO & 1 NC)

A2K1 - 3 Phase HV Contactor

The contactor is located in the upper right corner of the CPDU



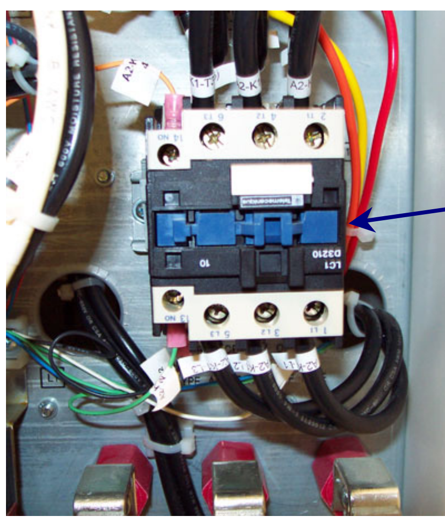
A2K1

Mounting of New Contactor - 2269902-11

CAUTION: Take necessary LOTO action when work on the CPDU

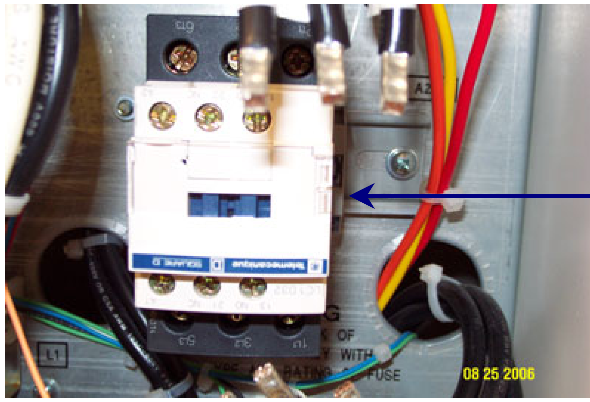
1. Disconnect wire leads from old contactor, noting wire labeling.

2. Remove old contactor from din rail, by releasing the spring-loaded clamp at the top of the base of the contactor. (*Note: Contactor is mounted upside down on din rail with respects to printing on contactor*).



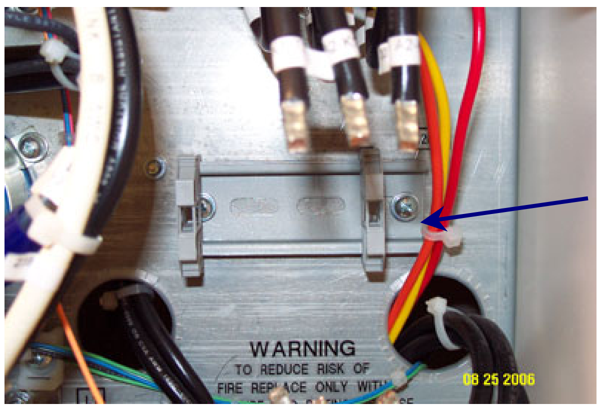
Note contactor is mounted upside down

3. Mount new contactor on din rail, upside down with respects to printing on contactor. Mount the contactor to the far left side of rail, to minimize distance required for wiring connections. (*Note: Mount the new contactor to the bottom edge of rail first and push upwards until the contactor snaps onto the top rail*.)



Note contactor is mounted upside down

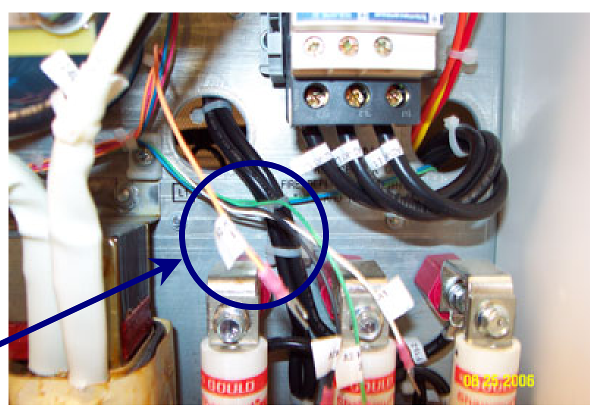
4. Make sure the two retention brackets are placed up against the contactor to keep it stable.



Contactor Retention Brackets

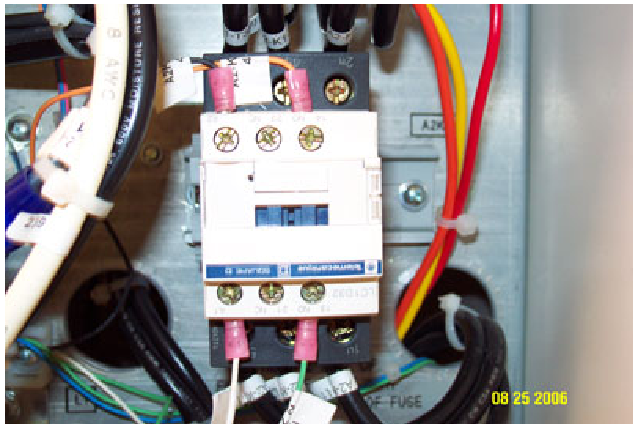
Connection of New Contactor - 2269902-11

1. Reconnect the 3 phase HV lines to the new contactor.



Note: In the next two steps, it may be necessary to remove tywraps and reroute the signal wires in order to reach their connection points.

2. Reconnect the two contactor coil wire leads to the A1 and A2 contacts.



3. Reconnect the two Aux. N.O. contacts wire leads to the N.O. contacts.
   (*Note: The N.C. contacts on new contactor are not used*.)

4. Re-check all connections are wired correctly referencing table below.

Signal	Wire Label	Contactor Label
HV	A2-K1-L_1	1 L_1
HV	A2-K1-L_2	3 L_2
HV	A2-K1-L_3	5 L_3
HV	A2-K1-T_1	2 T_1
HV	A2-K1-T_2	4 T_2
HV	A2-K1-T_3	6 T_3
120V Coil	A2-K1-A1	A1
120V Coil	A2-K1-A2	A2
Aux. Contact N.O.	A2-K1-13	13 N.O.
Aux. Contact N.O.	A2-K1-14	14 N.O.

5. Re-check all wiring connections are mounted securely, checking that wire lugs and ferrules are inserted fully and contact screws are tightened.

Michael L. Bauer
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